Deep-Learning-Based Continuous Multi-Subject Radar HAR Under Clutter

Abstract—

Keywords—

I. INTRODUCTION

Continuous, contactless monitoring of daily activities and basic physiological signals is increasingly important for ageing populations, post-operative recovery, and chronic disease management. Video systems raise persistent privacy concerns and are brittle under low light or occlusion, while wearables depend on user compliance, can be uncomfortable, and are often forgotten. Radar sensing offers an attractive alternative: it is illumination-independent, can penetrate clothing, and inherently protects appearance-level identity, making it well suited for long-term monitoring in homes and clinics[1].

Over the last decade, a substantial body of work has applied radio-frequency sensing to human activity recognition (HAR) and vital-sign monitoring using FMCW and CW radars. Micro-Doppler signatures extracted from range–Doppler or time–frequency representations have been used to classify gestures, falls, and coarse activities with convolutional and recurrent neural networks, often achieving high accuracy in controlled, single-subject environments, including recent continuous, multi-label and fusion-based HAR systems[2],[3]. Radar has also been shown to recover respiration and heartbeat by tracking tiny chest wall displacements with CW or narrowband systems and appropriate spectral processing[1]. These results demonstrate that RF sensing can capture rich kinematic and cardio-respiratory information without cameras or body-worn sensors.

II. LITERATURE REVIEW

Despite this progress, three gaps remain salient for real-world deployment in assisted-living and rehabilitation settings. First, **cross-environment robustness** is weak: models trained in one room or with one sensor placement often degrade sharply when moved to new layouts or levels of clutter, even when the nominal activities are unchanged[4]. Second, most datasets and models focus on **single-subject, isolated actions**, whereas actual homes contain multiple occupants, fans, and moving reflectors; multi-person radar HAR and vital-sign estimation remain comparatively under-explored, despite work on distributed or multi-radar setups[5]. Third, even when continuous recordings are collected, many systems still segment data into short, single-label windows for classification, sidestepping the harder problem of **continuous, multi-label timelines** with overlapping activities and evolving physiological state[2]. At the same time, privacy-sensitive activity datasets confirm that radar can be deployed in bathrooms and other sensitive spaces without exposing appearance-level identity [6]. Together, these gaps limit the reliability of current radar HAR systems outside of the lab.

From an evaluation perspective, recent radar HAR papers have moved beyond clip-level accuracy toward metrics defined on continuous timelines. [2] reports frame-wise and segment-wise performance for multi-label activity streams, emphasizing segmental F1 scores for overlapping actions and highlighting how naive windowing can underestimate fragmentation errors [5] and [7] setups likewise rely on ROC–AUC and segmental measures for detection-style tasks such as falls and rare events in continuous streams. This reflects a broader shift from closed-set classification toward detection and temporal segmentation, where missed onsets and fragmented segments matter; however, most published datasets still rely on carefully scripted routines, short recording durations, and relatively small subject pools.

Parallel lines of work point to additional opportunities and constraints that shape our problem. On the sensing side, phased-array and MIMO architectures, including phase-only beamforming, RIS-enabled array radars, and mainlobe-jamming suppression schemes, have shown that array-domain processing can suppress interference and enhance angular selectivity in dense environments[8],[9],[10]. Radar micro-Doppler and reflectometry have also been explored as proxies for muscle activation and hydration state, as covered in various works[11]-[12], but these applications are outside the scope of this project.

On the learning side, compact CNNs, temporal convolutional networks, and state-space models for radar spectrograms, including continuous multi-label HAR and multi-radar fusion models, have enabled inference with sub-millisecond latency and sub-megabyte footprints on embedded hardware, which is critical for edge deployment on platforms such as the Raspberry Pi or embedded SoCs [7],[2],[5]. Recent work has further specialized selective state-space models for radar micro-Doppler HAR, showing that Mamba-style architectures can achieve competitive accuracy with improved efficiency[13]. A good example is[2], which treats continuous HAR as a frame-wise multi-label problem over spectrogram sequences, using a CNN + Bi-GRU head to predict overlapping activities and showing that careful handling of temporal overlap and label smoothing is essential once actions are no longer isolated clips. [3] instead casts radar HAR as an image-like task and apply a multidomain fusion Vision Transformer that jointly attends over several time–frequency views, achieving strong accuracy but with tens of millions of parameters and quadratic self-attention cost. Complementary fusion work such as [7] shows that attention over multiple radar spectrogram "views" (e.g., different bands or vantage points) can significantly improve robustness in cross-environment settings, but still relies on comparatively heavy CNN/Transformer backbones.

Selective state-space models such as Mamba explicitly target this efficiency gap. [14] introduces Mamba as a linear-time sequence model with input-dependent state updates and a hardware-aware scan algorithm, matching or exceeding Transformer performance on language, audio, and genomics while using fewer FLOPs on long sequences [13] adapts this idea to radar micro-Doppler by proposing RadMamba, which combines a Chan-DS convolutional front-end, Doppler-aligned spectrogram segmentation, and a compact Mamba SSM head; across three datasets (DIAT, CI4R, UoG2020) RadMamba matches or surpasses prior CNN/ViT models while using only 6.7k parameters, about 1/400 of some Transformer baselines.

For a potential CN0566-based HAR system, these results suggest that a scaled-down RadMamba-style pipeline, Chan-DS-style spectral front-end, Doppler-aligned sequence construction, and a tiny Mamba/SSM backbone adapted to beamformed FMCW/CW modes, offers a promising compromise between continuous-sequence accuracy and edge-compute constraints.

III. Bibliography

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